MILITARY SPECIFICATION

CLOTH, RAYON
(For Liners For Recoilless Cartridge Cases)

1. SCOPE

1.1 This specification covers one type of rayon cloth for use in liners for recoilless cartridge cases.

2. APPLICABLE SPECIFICATIONS

2.1 The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-591 - Boxes, Wood-Cleated-Fiberboard.
NN-B-601 - Boxes, Wood-Cleated-Plywood, For Domestic Shipment.
NN-B-621 - Boxes, Wood, Nailed and Lock-Corner.
NN-B-631 - Boxes, Wood, Wirebound, For Domestic Shipment.
CCC-T-191 - Textiles; General Specifications, Test Methods.

MILITARY

MIL-B-130 - Barrier-Material, Paper, Noncorrosive.
MIL-G-2550- General Specification for Ammunition Except Small Arms Ammunition.
MIL-P-10025- Packing and Marking for Domestic Shipment of Inert Ammunition Components, General Specification for.

(Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. -The rayon cloth covered by this specification shall be woven from viscose rayon yarns, in both warp and filling. The rayon shall contain no acetyl groups and not more than 0.2 percent sulfur.

3.2 Color. - The color of the rayon cloth shall be natural unbleached or bleached white.

3.3 Appearance. - The rayon cloth shall be uniform and shall have no visible imperfections.

3.4 Width. - The width of the cloth shall be as specified (see 6.1). The specified width shall be not less than 36 inches. The width of the cloth shall be determined as specified in 4.4.1.

3.5 Weight. - The weight of the rayon cloth shall be between 5.00 and 5.75 ounces per square yard, determined as specified in 4.4.2.

3.6 Breaking strength. - The breaking strength of the cloth, determined as specified in 4.4.3, shall be as follows:

Warp direction: Minimum 70 pounds, Maximum 140 pounds.
Filling direction: Minimum 70 pounds, Maximum 140 pounds.

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3.7 Ether extract.- The ether extract of the rayon cloth shall be not more than 1.0 percent by weight, determined as specified in 4.4.4.

3.8 Acidity or alkalinity.- The acidity or alkalinity of the water extract of the rayon cloth, determined as specified in 4.4.5, shall be as follows:

Mineral - None
Organic - 0.10 percent, maximum

3.9 Ash.- The ash content of the cloth, determined as specified in 4.4.6, shall not be more than 0.2 percent by weight.

3.10 Air permeability.- The air permeability of the cloth, determined as specified in 4.4.7, shall be 50 ± 15 cubic feet of air per minute per square foot, at a pressure differential of 1/2 inch of water across the fabric.

3.11 Workmanship.- The finished cloth shall be uniformly and closely woven and shall contain no weighting materials. A plain weave is preferred, but other weaves may be employed provided they produce a fabric meeting the specified requirements.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot.- A lot shall consist of 100,000 square yards, maximum.

4.2 Sampling.- Ten percent of the containers, but in no case more than 10 or less than 3 containers, shall be selected by the inspector so as to be representative of the lot. If there are less than 4 containers in the lot, all the containers shall be sampled. Approximately 3 yards of material shall be taken from each selected container. Each of these primary samples shall be placed in a securely closed container, labeled so that the container from which it was taken can be identified. A composite sample consisting of approximately 1½ yards of each of the primary samples shall be made. The composite sample shall be placed in a securely closed container labeled to show the name of the material, name of manufacturer, plant, contract or order number, lot number, and lot size. All acceptance tests shall be made on the composite sample, using as many of the pieces as possible in making each of the tests. However, if it becomes apparent during sampling that the lot is not uniform, the inspector may require that any of the primary samples be tested for compliance with the specified requirements. All primary samples shall be held for possible future examination should the composite sample fail to meet the requirements.

4.3 Inspection.- Inspection shall be in accordance with Specification MIL-C-2550, and shall be at the point of delivery unless otherwise specified.

4.4 Laboratory tests.

4.4.1 Width.- Width of the cloth shall be determined as specified in Specification CCC-T-191.

4.4.2 Weight.- Weight of the cloth shall be determined as specified in Specification CCC-T-191.

4.4.3 Breaking strength.- Breaking strength of the cloth shall be determined by the grab method, as specified in Specification CCC-T-191.

4.4.4 Ether extract.- Cut approximately 0.5 gram of cloth from each of the pieces comprising the sample and weigh all the cuttings together. Transfer the cuttings to a Soxhlet extractor or other suitable extraction apparatus, having a tared receiving flask, and extract with ether for 1½ hours. Disconnect the receiving flask when most of the ether has collected in the middle section of the apparatus. Heat the receiving flask on a steam bath to evaporate any ether in the flask or accomplish this by means of a current of dry air. Heat the flask for 1 hour in an oven maintained at 100°C ± 2°C. Cool in a desiccator and weigh. Calculate the increase in weight of the flask to percent ether extract of the cloth.

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4.4.5 Acidity or alkalinity.- Cut approximately 0.5 gram of cloth from each of the pieces comprising the sample and weigh all the cuttings together. Transfer the cuttings to a 250-ml beaker, add 150 ml of neutral, freshly-boiled distilled water, and boil for 30 minutes. Keep the beaker covered with a watch glass during the boiling and do not allow the volume of the liquid at any time to become less than 75 ml. At the end of the 30 minute period add sufficient distilled water to bring the volume to 150 ml, and cool to room temperature. Add pieces of red and blue litmus paper which have been found to be sensitive to N/1000 sodium hydroxide and N/1000 sulphuric acid, respectively, when immersed in such solutions for 3 minutes. At the end of 3 minutes note if the extract of the sample is neutral, as indicated by the failure of either piece of litmus paper to be changed to the color produced by immersion of the paper in N/1000 acid or alkali. If the litmus paper is changed at the end of 3 minutes, filter the solution and titrate portions of the filtrate with approximately N/10 alkali or acid respectively, using phenolphthalein as the indicator. Calculate the acidity or alkalinity as percent acetic acid or sodium hydroxide, correspondingly.

4.4.6 Ash.- Cut small portions of cloth from each of the pieces comprising the sample until a total of approximately 2 grams of cloth is obtained. Weight all the cuttings together, transfer them to a tared platinum or porcelain crucible, and moisten them with concentrated nitric acid. Heat the crucible on a steam bath for 1 hour and then heat carefully over a flame so as to avoid any loss. When the contents of the crucible have been charred, heat to dull redness until all carbonaceous matter has been burned off. Cool the crucible in a desiccator and weigh. Calculate the gain in weight of the crucible to percent ash content of the cloth.

4.4.7 Air permeability.- Determine the air permeability of the cloth by the use of a self-contained instrument, a schematic diagram of which is illustrated on figure 2 of Specification CCC-T-191. Level the instrument by means of the leveling device mounted thereon. Fill the reservoir with water and replace the cap, making the seal practically air-tight. With both chambers open to the air adjust the scales back of the vertical and the inclined glass gages, so that the zero of the scale coincides with the bottom of the water meniscus in the respective gage. Cut the specimen of cloth in the form of an octagon, approximately 3 inches on a side, two sides of which run parallel to the warp and 2 sides parallel to the filling. Insert the specimen as shown on figure 2 of Specification CCC-T-191, fastening the clamps so that the long sides of two of the clamps will run parallel to the filling threads of the specimen. It is important that the specimen be clamped as described. Adjust the tension weights so that the fabric fits smoothly over the opening, with all portions of the cloth being as nearly as possible under the same tension. Clamp the hood in position, holding the specimen securely in place. By trial runs determine the most suitable air orifice for the fabric being tested. Choose an orifice which will give a water pressure reading on the vertical gage (which indicates the pressure drop across the orifice) of more than 2.0 inches but less than 11.81 inches, when 1/2 inch-of-water pressure is maintained across the fabric, as indicated on the inclined gage having a 1 to 10 slope. Start the suction fan motor, and adjust its speed by means of the rheostat until the inclined gage indicates 1/2 inch-of-water pressure. Record the reading shown on the vertical gage. Switch off the motor. The bottom of the menisci of the water columns of both the vertical and the inclined gages must return to their original positions; i.e., coincident with the zero on the scale behind the respective gage. The failure of the water columns to return to their original positions indicates that either the water system is clogged or that the gages are dirty. In such case clean the water system and the gages, and repeat the determination.

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Correct the vertical gage reading to 60°F and 30 inches-of-mercury pressure by means of the following formula:

$$h_o = \frac{17.27\ (h \times p)}{(458\ t)}$$

where:

h_o = corrected vertical pressure gage reading, in inches-of-water at 60°F and 30 inches-of-mercury pressure.
h = observed vertical pressure gage reading, in inches of water.
p = prevailing barometric pressure, in inches of mercury.
t = prevailing temperature, in degrees Fahrenheit.

Calibrate the volume of air flowing through the particular orifice used in the instrument by means of the following air-flow calibration values for an orifice having a diameter of 1.0 mm (at 60°F and 30 inches of mercury pressure):

Discharge for 1.0 mm orifice

Corrected vertical gage reading (h_o) (Inches of water)	Air flow rate (liters per minute)
11.81	3.24
7.87	2.66
5.91	2.31
3.94	1.89
2.76	1.58

(Intermediate points shall be determined by interpolation.)

At a given pressure drop across an orifice, the volume of air passing through the orifice varies in direct proportion to the square of the diameter of the orifice. Thus, if there is a pressure drop of 2.76 inches across an 8.0 mm orifice, the computed air flow would be, according to the above calibration, $1.58 \times 8^2 = 101.1$ liters of air per minute. However, inasmuch as orifices cannot be constructed perfectly, the air-flow calibrations shall be as established by a National Bureau of Standards calibration of the actual orifice used. From the NBS calibration, calculate the liters of air passing per minute through the definite exposed area of the fabric. Translate the volume of air passing through the sample into terms of cubic feet per minute, and then into cubic feet per minute per square foot of cloth, in direct proportion to the definite exposed area of the test specimen (see 6.2).

4.5 Resubmission and retest.- If the composite sample, or any primary sample, fails to pass the specified tests, the lot shall be rejected. The contractor shall have the option of having a complete or a partial analysis made on samples, from any or all of the bales or boxes in the lot, at no expense to the Government. The contractor may then remove the defective portions of the lot and resubmit the lot for acceptance, provided that replacement of the defective portions is made to the satisfaction of the inspector. The resubmitted lot shall be accepted if the new samples, selected as specified in 4.2, pass all the required tests.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Unless otherwise specified, the rayon cloth shall be packaged in bolts, the cloth being either folded or rolled. If rolled, the cloth shall be wound on a cardboard or other suitable tube having an internal diameter of approximately 2 inches, its length approximating the width of the cloth, and of sufficient

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strength to prevent its distortion in shipping. No bolt shall contain any piece of cloth less than 20 yards in length. The bolt shall be wrapped in 3 or more thicknesses of paper conforming to type 1 of Specification MIL-B-130, except that the paper shall contain no substance that could transfer from the paper to the cloth. The wrapping shall be securely held by running three tape bands completely around the roll, and one strip of tape the full length of the overlap seam. The paper shall be long enough to tuck into the ends of the tube. The tape shall be gummed-kraft having a minimum width of 2½ inches, and 60-pound basis weight.

5.2 Packing. - Unless otherwise specified, cloth packaged as specified in 5.1 shall be packed in bales or in boxes. If packed in bales, the bales shall be wrapped in burlap. If packed in boxes, the boxes shall conform to Specification NN-B-591, NN-B-601, NN-B-621, or NN-B-631. The gross weight of the packed bale or box shall be not more than 200 pounds.

5.3 Marking.

5.3.1 Individual bolts. - Unless otherwise specified, a tag shall attached to each bolt of cloth indicating the width of the cloth, number of yards of cloth in the bolt, date of manufacture, name of manufacturer, and number of contract or order.

5.3.2 Shipping containers. - In addition to any special markings specified by the procuring activity (see 6.1) the shipping containers shall be marked in accordance with Specification MIL-P-10025.

6. NOTES

6.1 Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Width of cloth required (see 3.4).
c. Special markings of shipping containers required (see 5.3.2)

6.2 The air permeability of the cloth may be determined with the instrument described in National Bureau of Standards Research Paper R.P. 1471 entitled "Improved Instrument for Measuring the Air Permeability of Fabrics", Journal of Research of National Bureau of Standards Vol. 28, dated May 28, 1942; or by any other suitable instrument approved by the procuring agency.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with consideration on account of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1426 REPLACES NAVSHIPS FORM 4883, WHICH IS OBSOLETE C-0278

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